

Git Hands-On Lab 3 - Branching and Merging

Objective

To create a new Git branch, perform changes in the branch, commit the changes, merge the branch into the master branch, verify the commit history, and delete the merged branch.

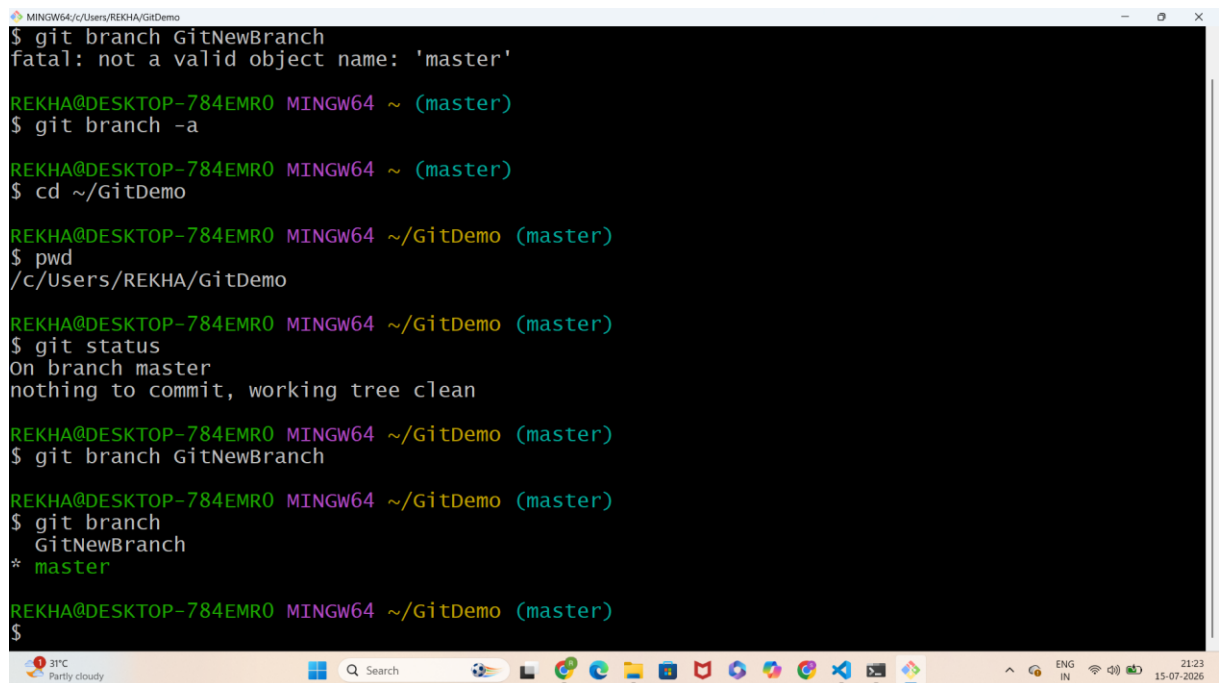
Step 1: Create a New Branch

Command:

```
git branch GitNewBranch
git branch
```

Observed Output:

```
GitNewBranch
* master
```



```
MINGW64/c/Users/REKHA/GitDemo
$ git branch GitNewBranch
fatal: not a valid object name: 'master'

REKHA@DESKTOP-784EMR0 MINGW64 ~ (master)
$ git branch -a

REKHA@DESKTOP-784EMR0 MINGW64 ~ (master)
$ cd ~/GitDemo

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ pwd
/c/Users/REKHA/GitDemo

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch GitNewBranch

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch
  GitNewBranch
* master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$
```

Step 2: Switch to the New Branch

Command:

```
git checkout GitNewBranch
git branch
```

Observed Output:

```
Switched to branch 'GitNewBranch'
* GitNewBranch
master
```

```
MINGW64/c/Users/REKHA/GitDemo
wBranch)
$ git branch
* GitNewBranch
  master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This file is created in GitNewBranch. " > branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ ls
app.log  branch.txt  log/  welcome.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        branch.txt

nothing added to commit but untracked files present (use "git add" to track)

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch 6e823d7] Added branch.txt in GitNewBranch
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$
```

Step 3: Create a File in the Branch

Commands:

```
echo "This file is created in GitNewBranch." > branch.txt
ls
git status
```

Observed Output:

branch.txt created successfully.
branch.txt displayed as an untracked file.

```
MINGW64/c/Users/REKHA/GitDemo
wBranch)
$ git branch
* GitNewBranch
  master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This file is created in GitNewBranch. " > branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ ls
app.log  branch.txt  log/  welcome.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        branch.txt

nothing added to commit but untracked files present (use "git add" to track)

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch 6e823d7] Added branch.txt in GitNewBranch
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$
```

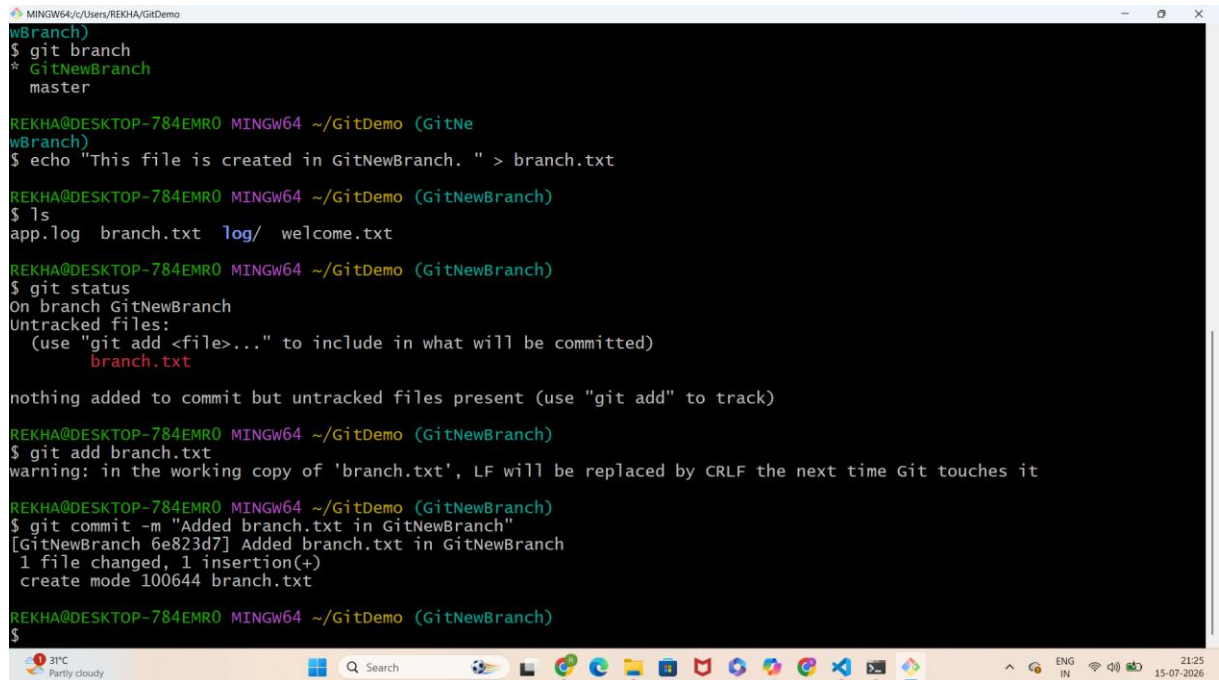
Step 4: Stage and Commit the File

Commands:

```
git add branch.txt
git commit -m "Added branch.txt in GitNewBranch"
```

Observed Output:

```
[GitNewBranch] Added branch.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branch.txt
```



The screenshot shows a Windows terminal window with the title bar 'MINGW64/c:/Users/REKHA/GitDemo'. The terminal content is as follows:

```
wBranch)
$ git branch
* GitNewBranch
  master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This file is created in GitNewBranch. " > branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ ls
app.log  branch.txt  log/  welcome.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    branch.txt

nothing added to commit but untracked files present (use "git add" to track)

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch 6e823d7] Added branch.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$
```

The terminal window has a taskbar at the bottom showing the Windows Start button, a search bar, and various application icons. The system tray on the right shows the date and time as 21:25 on 15-07-2026.

Step 5: Switch Back to Master

Command:

```
git checkout master
```

Observed Output:

```
Switched to branch 'master'
```

```
MINGW64/c/Users/REKHA/GitDemo
REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..cc84a8c
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch.

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 047e05e..6e823d7
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 6e823d7 (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch
* 047e05e Added .gitignore file
* 190ed62 Initial commit

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 6e823d7).

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch
* master

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$
```

Step 6: Compare Master and Branch

Command:

```
git diff master GitNewBranch
```

Observed Output:

Shows the differences between master and GitNewBranch, including the newly created branch.txt file.

```
MINGW64/c/Users/REKHA/GitDemo
REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..cc84a8c
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch.

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 047e05e..6e823d7
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 6e823d7 (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch
* 047e05e Added .gitignore file
* 190ed62 Initial commit

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 6e823d7).

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch
* master

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$
```

Step 7: Merge the Branch

Command:

```
git merge GitNewBranch
```

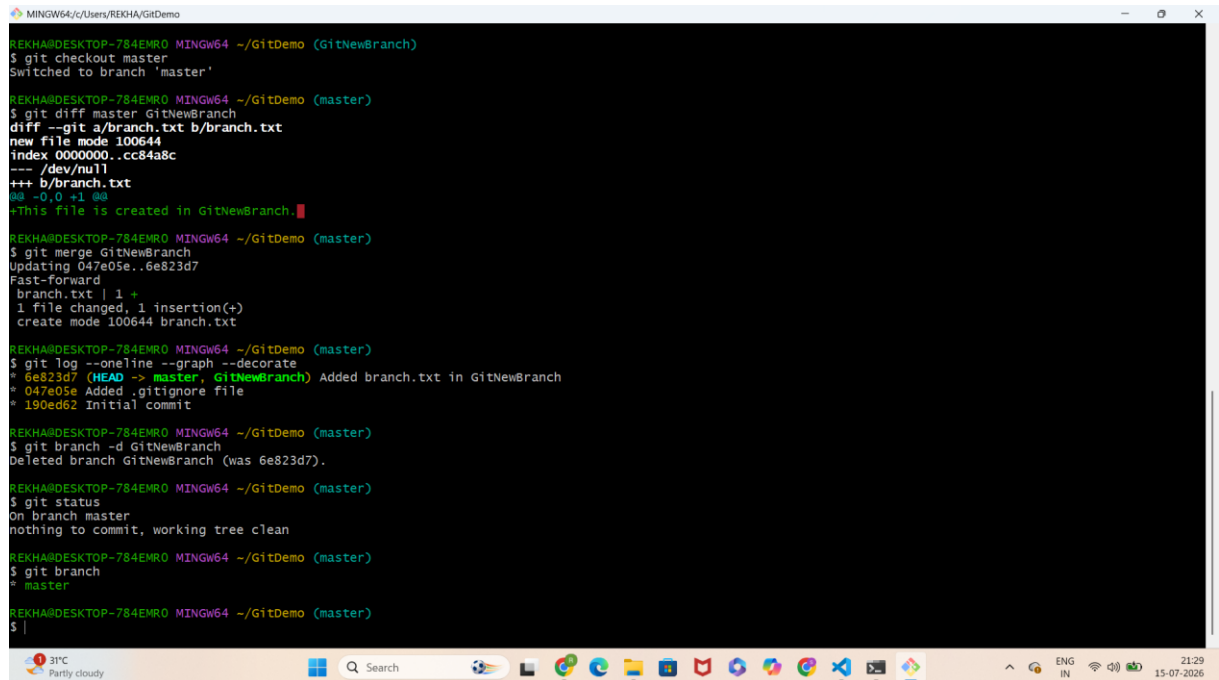
Observed Output:

Fast-forward

branch.txt | 1 +

1 file changed, 1 insertion(+)

create mode 100644 branch.txt



```
MINGW64/c/Users/REKHA/GitDemo
REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..cc84a8c
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch.

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 047e05e..6e823d7
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 6e823d7 (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch
* 047e05e Added .gitignore file
* 190ed62 Initial commit

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 6e823d7).

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch
* master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$
```

Step 8: View Commit History

Command:

```
git log --oneline --graph --decorate
```

Observed Output:

Displays the commit graph showing:

- Initial commit
- .gitignore commit
- branch.txt commit

```
MINGW64/c/Users/REKHA/GitDemo
REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..cc84a8c
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch.

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 047e05e..6e823d7
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 6e823d7 (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch
* 047e05e Added .gitignore file
* 190ed62 Initial commit

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 6e823d7).

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch
* master

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$
```

Step 9: Delete the Branch

Command:

```
git branch -d GitNewBranch
```

Observed Output:

Deleted branch GitNewBranch.

```
MINGW64/c/Users/REKHA/GitDemo
REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..cc84a8c
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch.

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 047e05e..6e823d7
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 6e823d7 (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch
* 047e05e Added .gitignore file
* 190ed62 Initial commit

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 6e823d7).

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$ git branch
* master

REKHA@DESKTOP-784EMRO MINGW64 ~/GitDemo (master)
$
```

Step 10: Verify Repository Status

Commands:

```
git status
```

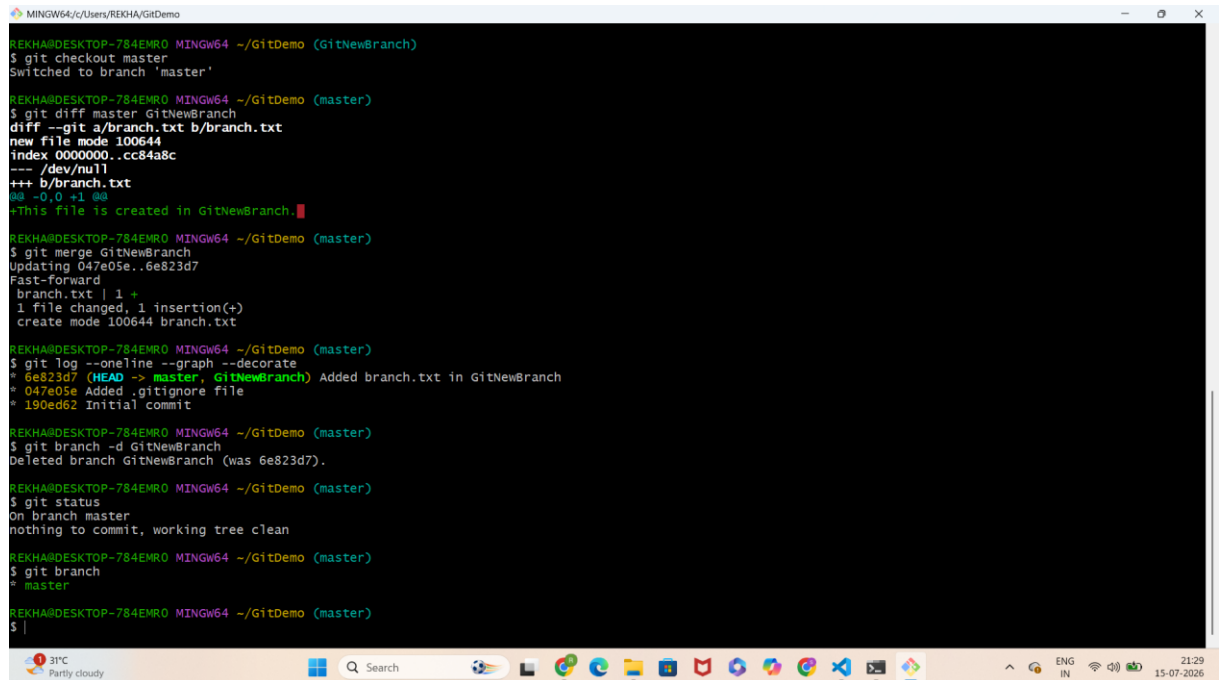
```
git branch
```

Observed Output:

On branch master

nothing to commit, working tree clean

Only master branch remains.



```
MINGW64/c/Users/REKHA/GitDemo
REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..cc84a8c
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch.

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating 047e05e..6e823d7
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 6e823d7 (HEAD -> master, GitNewBranch) Added branch.txt in GitNewBranch
* 047e05e Added .gitignore file
* 190ed62 Initial commit

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 6e823d7).

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$ git branch
* master

REKHA@DESKTOP-784EMR0 MINGW64 ~/GitDemo (master)
$
```

Result

The branching and merging operations were completed successfully. A new branch was created, changes were committed, merged into the master branch, the commit history was verified, the merged branch was deleted, and the repository was confirmed to be in a clean state.